

[ISOM2010] INTRODUCTION TO INFORMATION SYSTEMS

DIGITAL ECONOMY II

ENTERPRISE SYSTEMS

S3

PROF. KIM, YONGSUK

VARIOUS DEPARTMENTS IN A COMPANY



IN THE PRE-ERP ERA

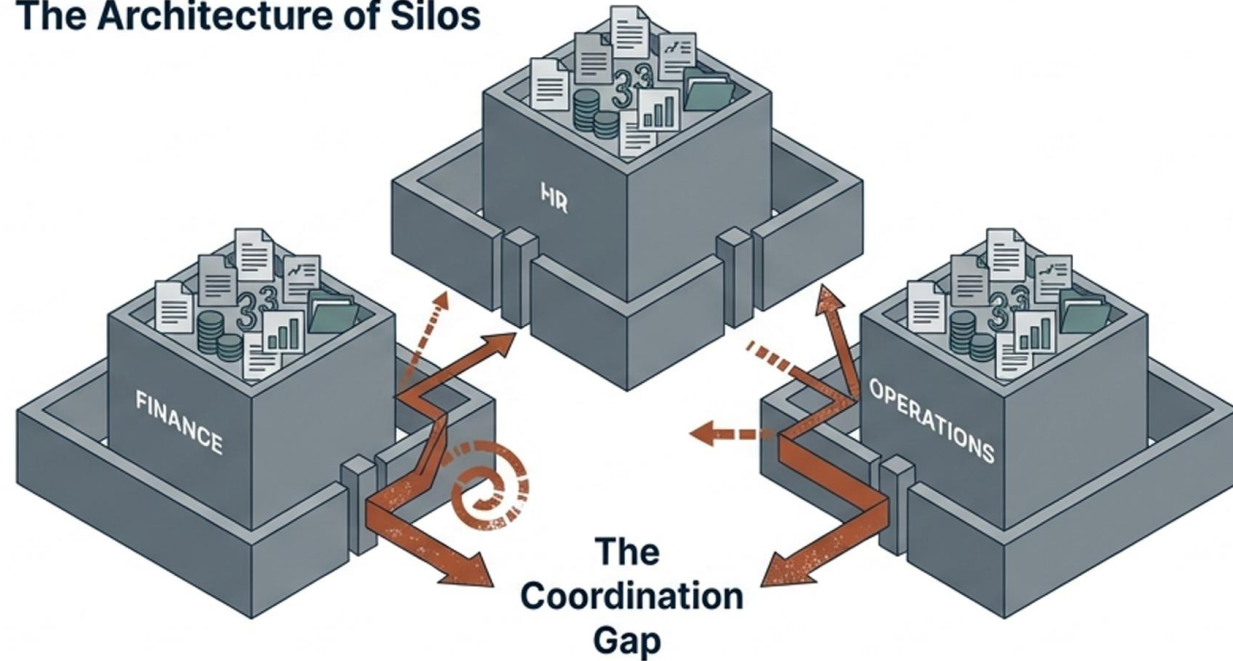
The PC Revolution (1980s)

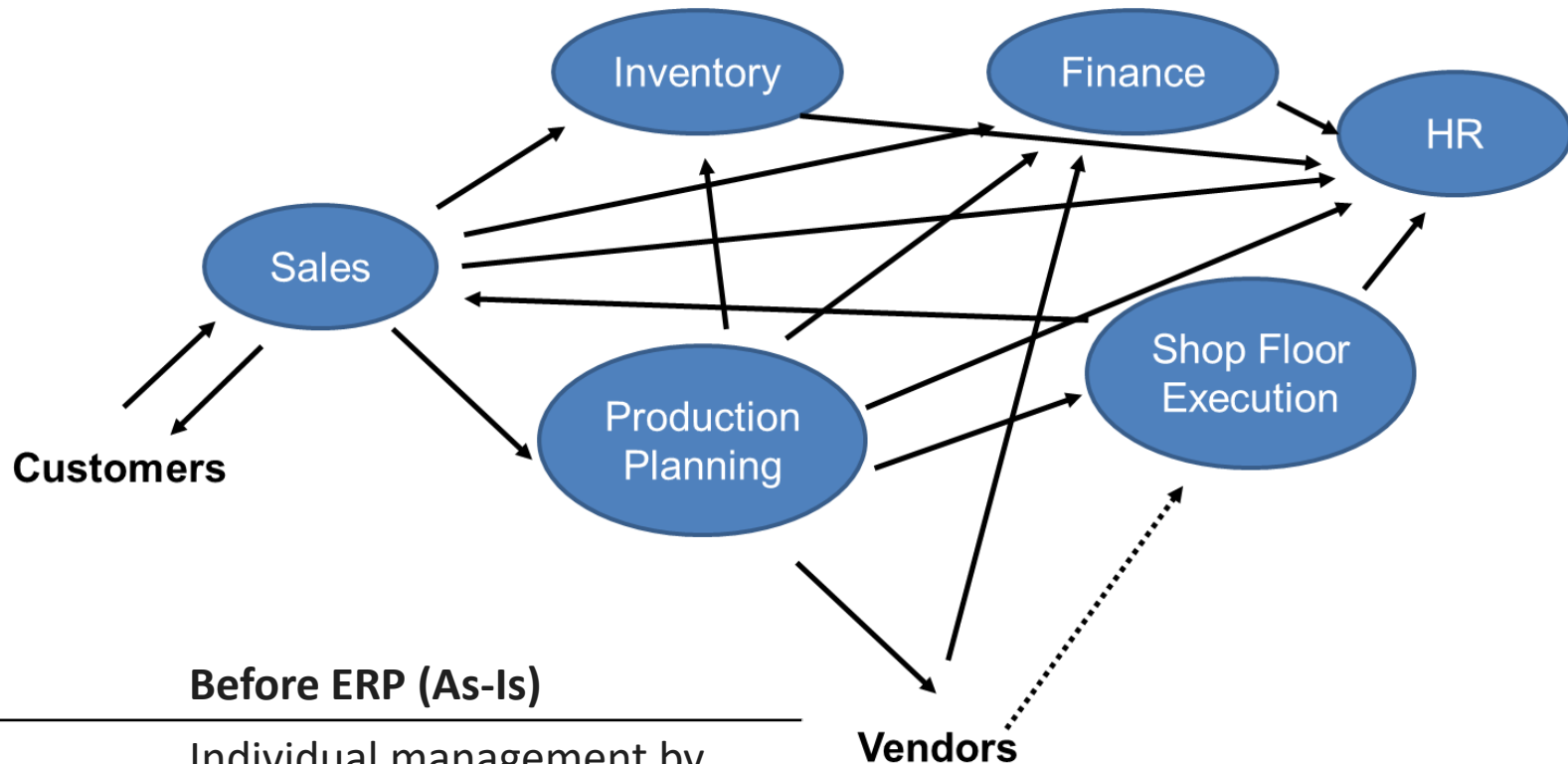
While personal computers automated specific tasks, they trapped data within departments. This created "Information Silos" where systems could not communicate.

Key Pain Points:

- Duplicate data entry across functions
- Inconsistent numbers (Finance vs. Sales)
- Reliance on manual spreadsheets
- Slow, reactive decision-making

The Architecture of Silos





Category

Before ERP (As-Is)

Data Management

Individual management by department (Data Silos)

Work Process

Manual data re-entry and transmission

Accuracy

Frequent data inconsistency between departments

Speed

Delays due to manual cross-departmental checks

PROBLEMS WITH DECENTRALIZED SYSTEMS

- Numerous disparate information systems are developed individually and locally over time.
- Integrating the data is time and money consuming.
- Inconsistencies and duplication of data
- Lack of timely information leads to customer dissatisfaction, loss of revenue and reputation.
- High inventory, material, and human resource costs

SOLUTION?

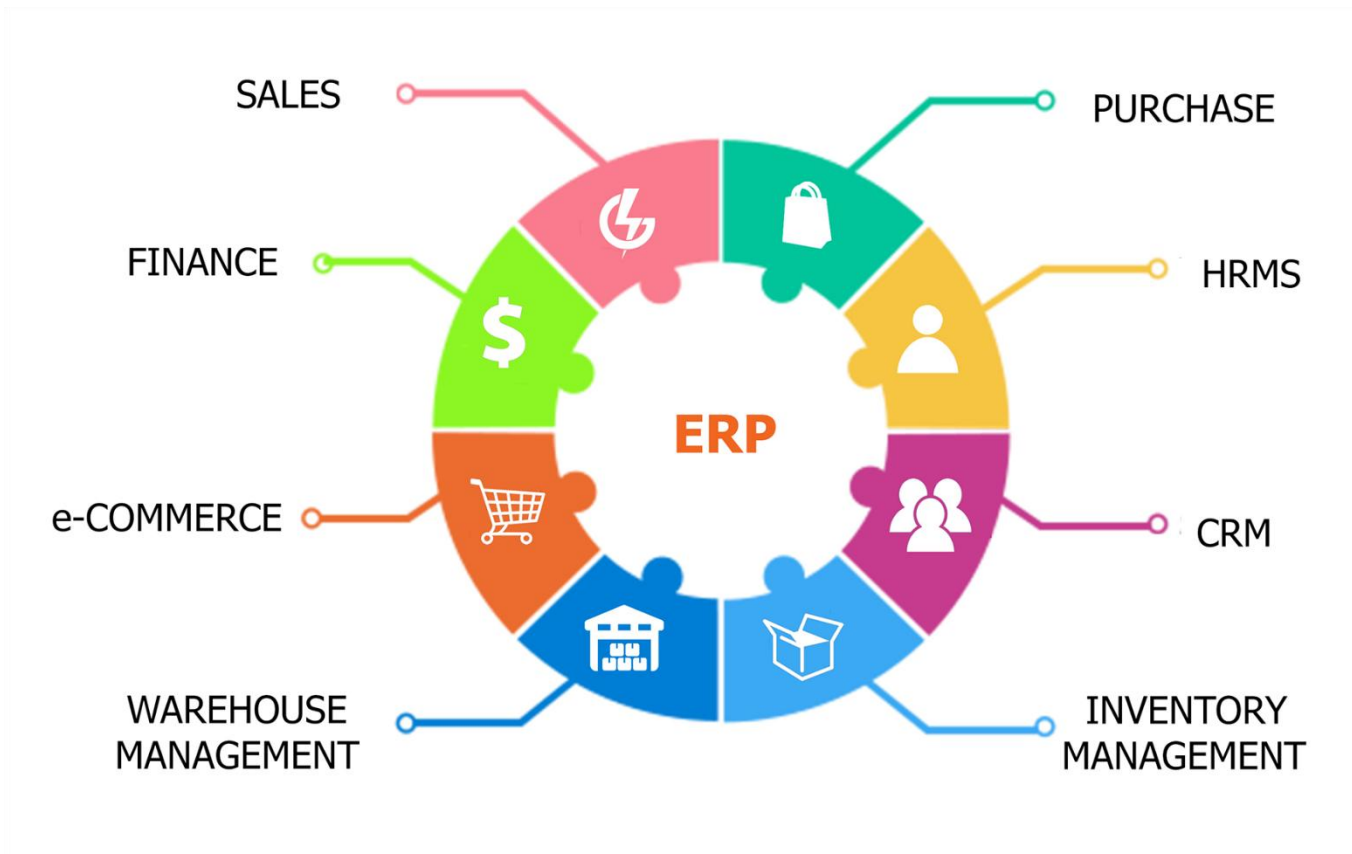
A centralized system (e.g., ERP)

ENTERPRISE SYSTEMS



>> ENTERPRISE RESOURCE PLANNING (ERP)

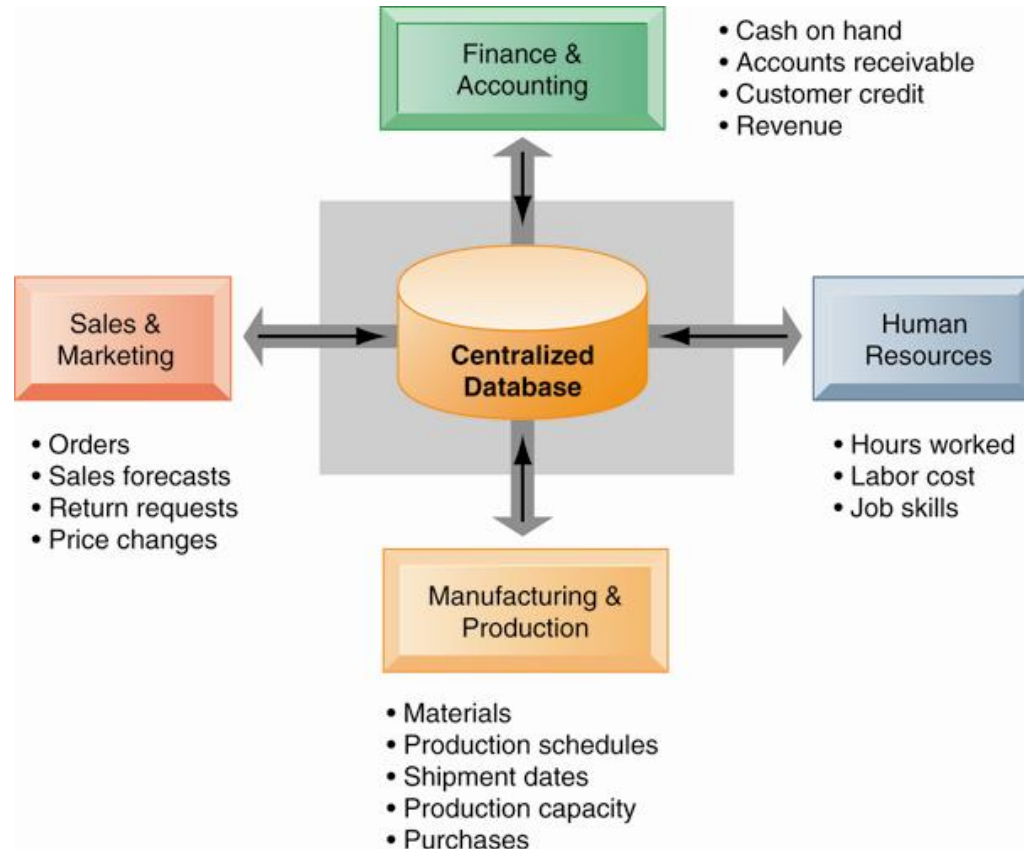
- An integrated software system to facilitate the flow of information *between all business functions in an organization*



So that they share a common database and information flow.

ENTERPRISE RESOURCE PLANNING (ERP)

ERP system features **a set of integrated software modules and a central database** that enables data to be **shared by many different business processes and functional areas** throughout the enterprise



HKUST INTRANET (ERP)

 Search with keywords, e.g. "ePDR", "Leave", "Expense", "Payslip", "Form", etc.



My Administrative
Corner

My Personal
Corner

Policies and
Guidelines

Service Requests and
Forms

Administrative
Systems

Analytics

FAQ

Student Related

- Student Information System 
- Faculty and Advisor Center 
- Class Enrollment Request 
- Textbook Services 
- RPG Professional Dev Courses 
- Research Output System 
- RPG Progress Report Approval 
- Thesis Examination System 

Finance Related

- Account Code Mapping Enquiry 
- General Budget Enquiry (HKUST) 
- Financial Status Enquiry for Projects (HKUST) 
- General Budget Enquiry (Others) 
- Financial Status Enquiry for Projects (Others) 
- Budget Requisition Enquiry 
- Centralized Approval and Delegation System 

Others

- Estates Inventory Management System 
- Conference Lodge Booking System 
- Learning Management System 
- Direct Marketing Opt-out Service 
- Rooms & Facilities Booking Portal 
- Shaw Auditorium Venue Application 
- Meeting Room @ Centralized Booking System 
- Communal Area @ Centralized Booking 

HKUST INTRANET (ERP)

Purchasing Related

- Requisition and Receiving Enquiry ★
- eQuotation System ★
- eProcurement / Requisition ★
- Purchasing Card System ★
- Goods Receiving ★
- Vendor & Product Enquiry ★

Extra Load Application and Payment System ★

Fixed Asset Web Portal ★

Fixed Assets for Departmental Transfer ★

Non-Fixed Assets for Departmental Transfer ★

Finance Portal ★

Travel Portal ★

Safety Related

- H&S Training & Exam Application ★
- Hazardous & Chemical Inventory ★
- Chemical Waste Collection ★
- Radioactive Waste Collection ★

Human Resource Related

Leave Enquiry ★

Leave Summary Report ★

Leave Endorsement / Approval ★

HKUST INTRANET (ERP)

Student Related

UG Admissions (2026-27 Entry)	★
UG Admissions (2025-26 Entry)	★
UG Admissions (2024-25 Entry)	★
PG Admissions (2026-27 Entry)	★
PG Admissions (2025-26 Entry)	★
PG Admissions (2024-25 Entry)	★
Student visa / entry permit administration	★
HKUST Card administration (for students)	★
UG Credit Transfer Administration	★
RPG professional development courses administration	★

Student Service Related

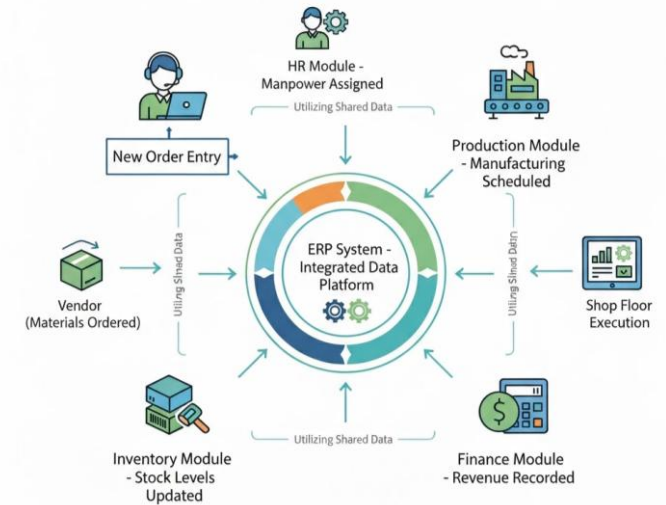
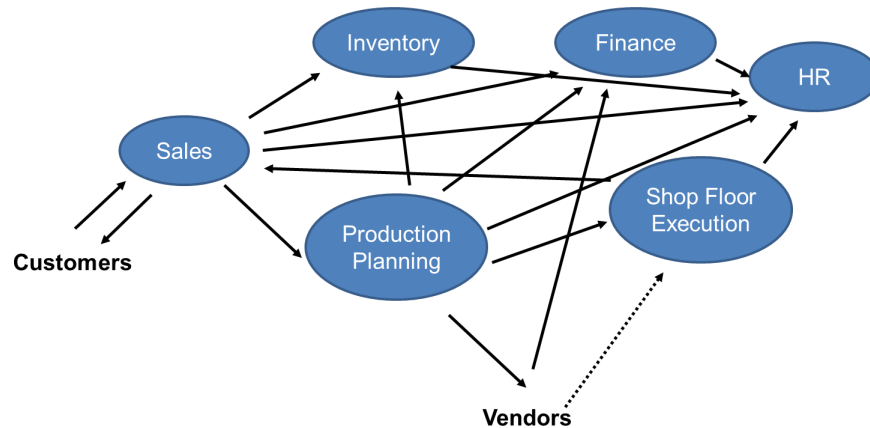
Hall management	★
Hall guest services administration	★
Healthy lifestyle course	★
Storage locker administration	★
Textbook services administration	★
Graduation services	★
Diploma mailing management	★

Finance Related

Financial and Purchasing Management	★
Funds / Project Accounting System	★

Others

Facilities / Amenities User Management System	★
Staff attendance records and administration (CMO)	★
Staff Quarters System (CMO)	★
Staff attendance records and administration (DSTO)	★
Room booking administration	★
Estates Inventory Management System	★
Alumni Voting System	★
Conference Lodge Booking System administration	★
Vendor & Product administration	★



Category	Before ERP (As-Is)	After ERP (To-Be)
Data Management	Individual management by department (Data Silos)	Centralized Shared Database
Work Process	Manual data re-entry and transmission	Real-time Automatic Updates
Accuracy	Frequent data inconsistency between departments	Single Source of Truth (SSOT)
Speed	Delays due to manual cross-departmental checks	Immediate sync of inventory, finance, and production

SHOP FLOOR EXECUTION TO CENTRAL ERP

In an integrated ERP environment, the "Shop Floor" is no longer an isolated island. It both consumes and produces critical real-time data that synchronizes the entire supply chain.

Data Type	Flow Direction	Impact on Other Modules
Production Status	Shop Floor → ERP	Sales: Provides accurate "Available to Promise" (ATP) dates for customers.
Material Usage	Shop Floor → ERP	Inventory: Automatically deducts raw materials and triggers reorders from Vendors.
Labor Hours	Shop Floor → ERP	HR: Records actual manpower assigned for payroll and productivity analysis.
Output / Finished Goods	Shop Floor → ERP	Finance: Calculates the "Cost of Goods Sold" (COGS) and updates asset value.

ERP BENEFITS AND CHALLENGES

➤ *ERP Benefits*

- **Improve business process efficiency**
 - Standardize and coordinate business processes
- **Reduce data entry redundancy**
- **Improve information flow and sharing within organization**
- **Improve organization's response to environment**
- **Analytical tools to evaluate overall organizational performance**

➤ *ERP Challenges*

- **Cost**
- **Resistance to changes**
- **Map business processes to software processes**
- **Cross-department integration**
- **Legacy Systems (= existing old systems)**

: ORGANIZATIONAL TRANSFORMATION, NOT JUST SW INSTALLMENT

The core challenge is cultural adaptation and process redesign, not technical deployment.



The Software (Easy)



The Organization (Hard)

Process Discipline

Employees must adapt workflows to standardized "best practices" embedded in the software. Customization is costly; adaptation is necessary.

Job Redesign

Roles change significantly. Manual data entry is replaced by analysis and process management.

The Hidden Cost

Training and change management often consume more resources than the technology license itself.



Insight: Most ERP failures are **organizational, not technical**. Success requires deep cultural commitment to standardized processes.

ENTERPRISE SYSTEMS AND VALUE CHAIN



ORACLE



Apple dismisses rumors about order cuts as iPad and iPhone shipments soar

CEO Tim Cook says Mac and iPad Mini sales would have been better had it not been for **supply-chain constraints**

By Ted Samson | InfoWorld

 Follow @tsamson_IW

 digitimes

[Memory capacity shortage sparks 2026 supply chain scramble, raising disruption risks](#)

The memory industry faces a severe capacity crunch as global manufacturers rush **upstream** to secure DRAM production for 2026, sparking fears...

20 Oct 2025

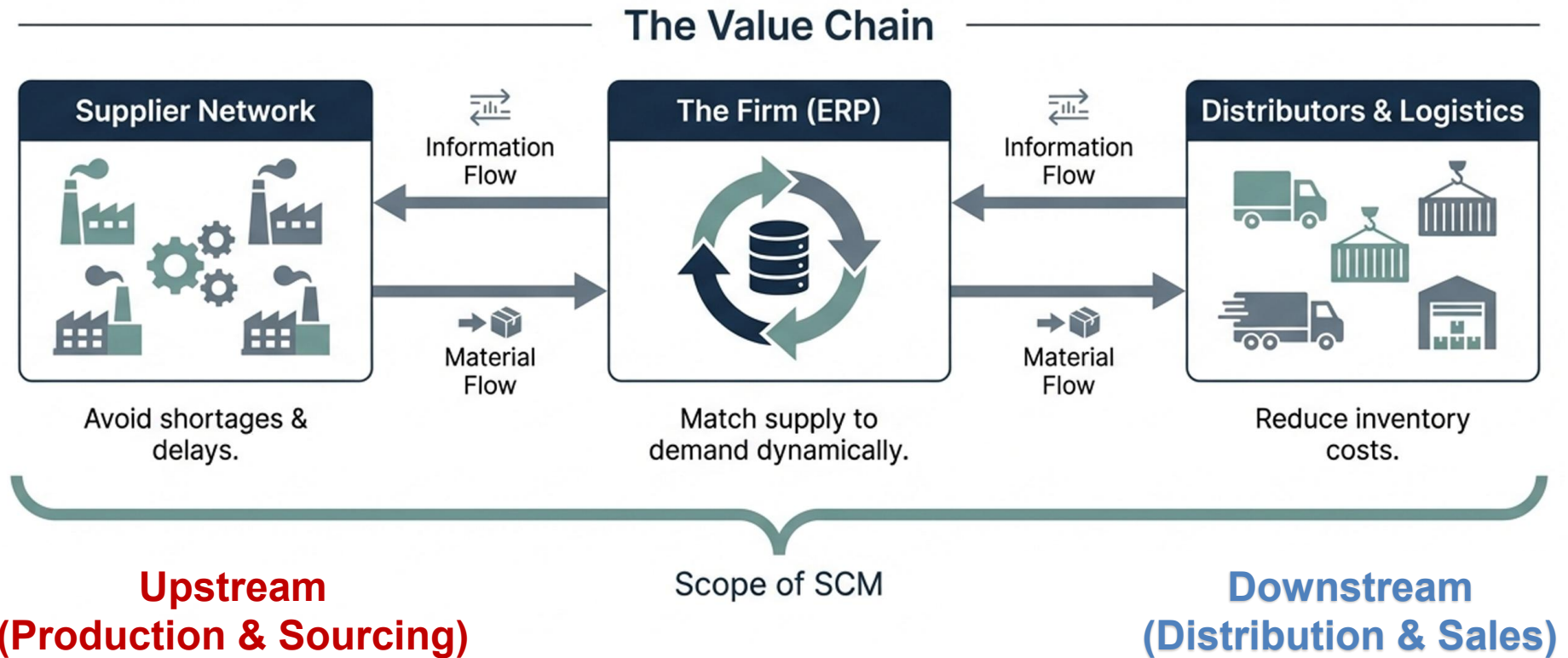
 China Briefing

How Will China's Rare Earth Export Controls Impact Industries and Businesses?

China's rare earth export controls have expanded to cover more products, affecting **global supply chains** and critical industries.

10 Nov 2025

Supply Chain Management Extends Coordination Beyond the Firm



>> SUPPLY CHAIN

- a **network** between a company and its suppliers to **produce** and **distribute** a specific product
 - Encompasses upstream, internal (midstream), and downstream activities



Visit: <https://www.youtube.com/watch?v=UBSOiHUctrY>

COCA COLA'S SUPPLY CHAIN



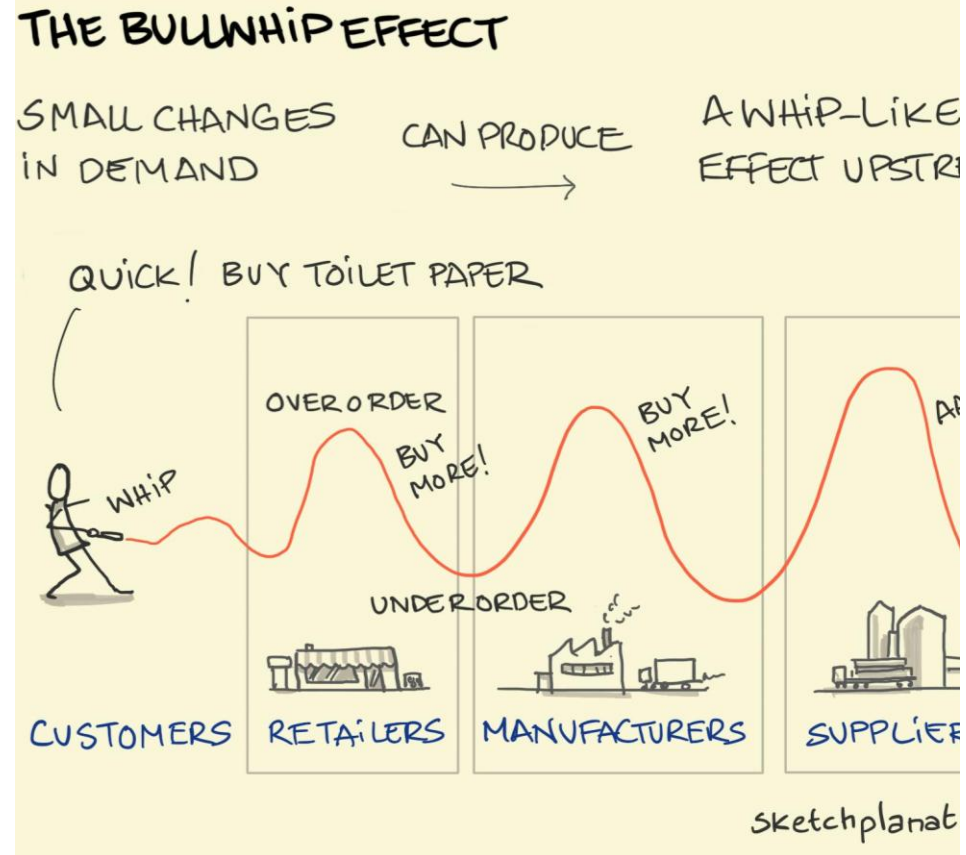
>> SUPPLY CHAIN

- The Core SCM Conflict: Efficiency vs. Responsiveness
 - The **Bullwhip Effect** forces a supply chain to choose between two "evils" if it isn't managed correctly:
 - **Excess Inventory (Inefficiency)**: Holding too much stock "just in case" (to avoid the whip's snap), which ties up cash and leads to waste.
 - **Stockouts (Unresponsiveness)**: Not having enough product when the "snap" happens, leading to lost revenue and angry customers.
- An *optimal* supply chain results in lower costs and a faster production cycle.
 - Supply chain management (SCM)
(the management of supply chains)

BULLWHIP EFFECT IN SUPPLY CHAIN

The bullwhip effect is the amplification of demand variability up the supply chain, where small changes in consumer demand lead to progressively larger order fluctuations and excess inventory for manufacturers and distributors.

It is caused by demand variability, poor communication and long lead (waiting) times.



>> SUPPLY CHAIN MANAGEMENT (SCM)

- Coordinates and synchronizes activities of all stakeholders* in the supply chain.
- Goal is to move the correct amount of products from source to point of consumption **as quickly as possible** and **at the lowest cost**
- Technology enables seamless and efficient exchanges of information, goods, and services among stakeholders.

* Stakeholders: suppliers, manufacturers, distributors, dealers, etc.

SCM BENEFITS AND CHALLENGES

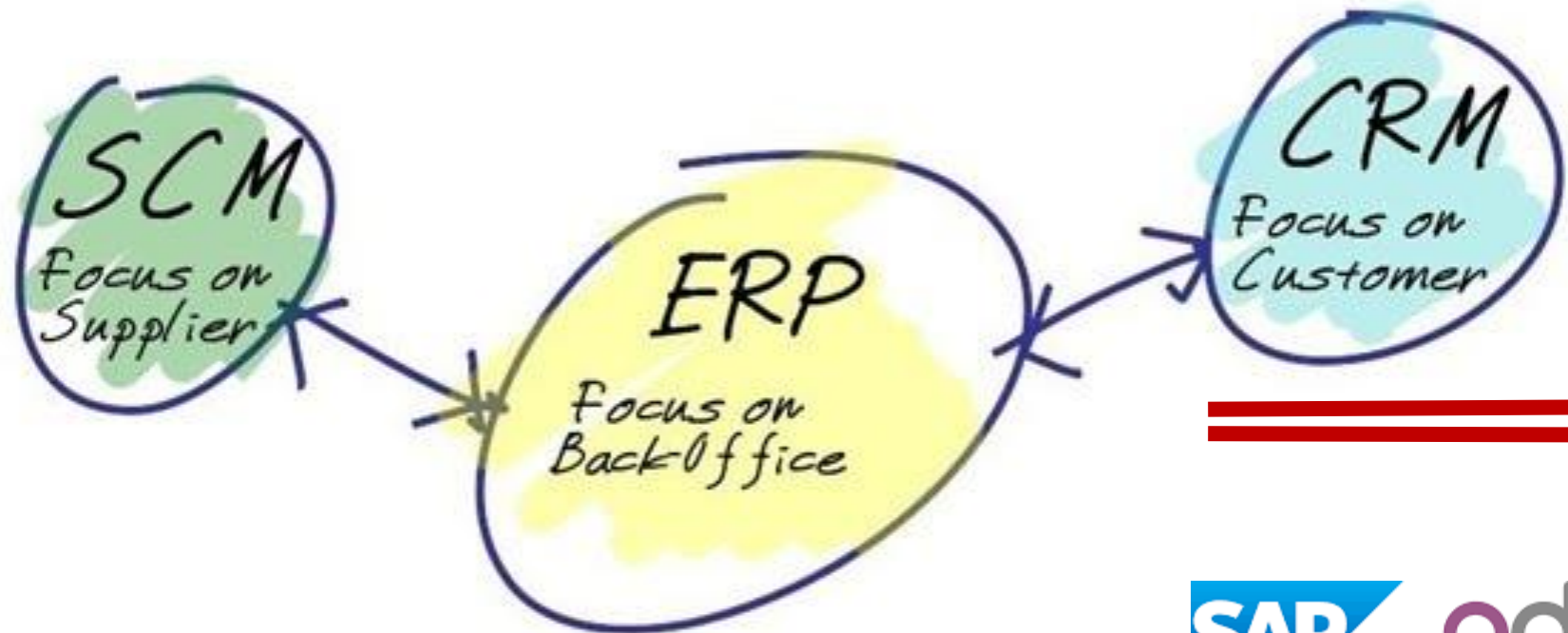
➤ *SCM Benefits*

- **Greater visibility of supply chain**
- **Lower inventory holding**
- **Reduce stock outs**
- **Lower lead time*** (*the time between the initiation and completion of a production process)
- **Tighter relationship with partners**
- **Higher productivity**

➤ *SCM Challenges*

- **Getting suppliers/partners to come on board**
 - Costs
 - Trust between business partners
- **Internal resistance to change**
- **Learning curve (much time/effort needed to learn)**

ENTERPRISE SYSTEMS AND VALUE CHAIN

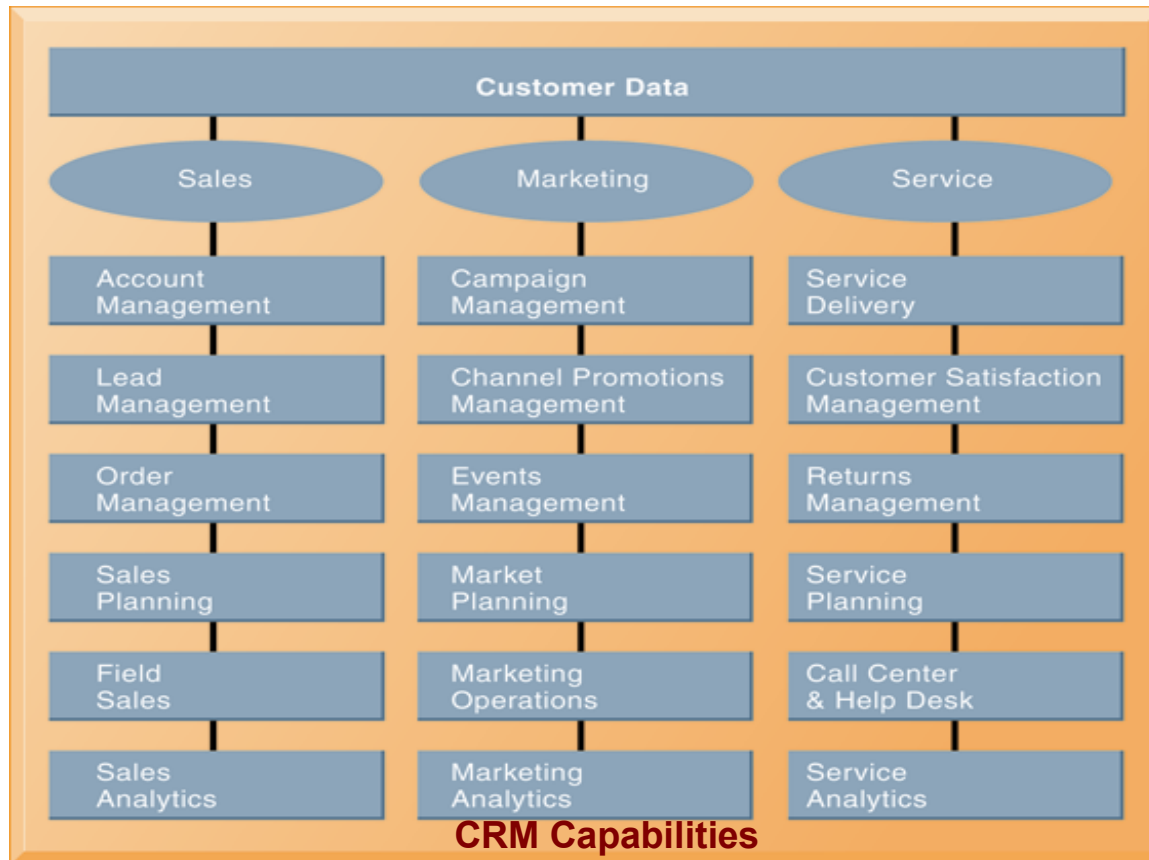


ORACLE



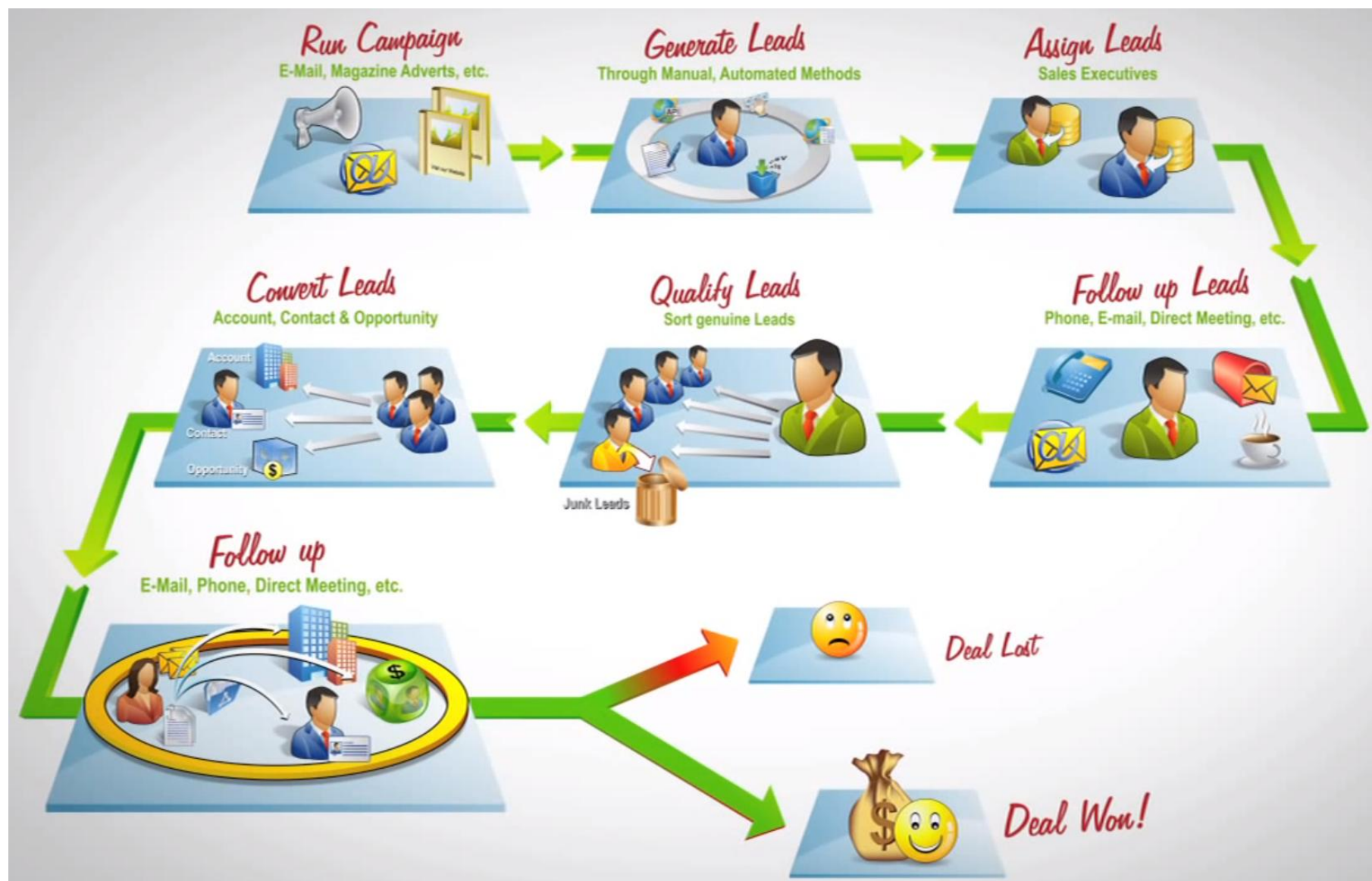
>> Customer Relationship Management (CRM)

- A customer service approach that focuses on **building long-term and sustainable customer relationship that add value** for customers and the company.



- Capture and integrate **customer data** from all over the organization
- Consolidate and analyze customer data
- Provide single enterprise view of customers
- Help manage relationship with customers

CRM SOFTWARE FEATURES





CRM BENEFITS AND CHALLENGES

▣ *CRM Benefits*

- Identify most profitable customers
- Improve customer satisfaction
- Increase customer retention
- Increase sales

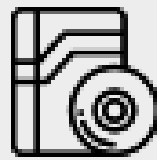
▣ *CRM Challenges*

- Collecting and analyzing customer information
- Too much data and too many channels (touch points)
- Enhancing customer experience
- Sales - marketing – service process integration (or automation)

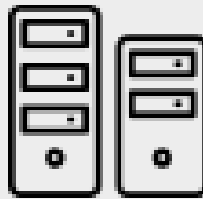
CLOUD COMPUTING



Virtual
Desktop



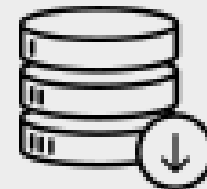
Software
platform



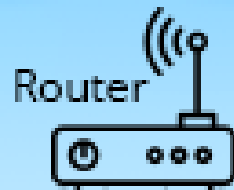
Servers



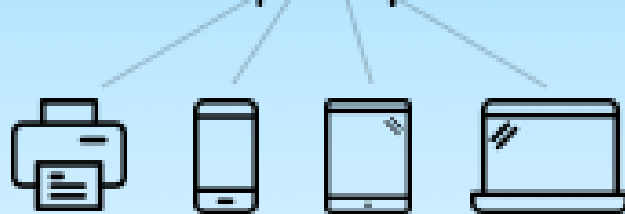
Applications



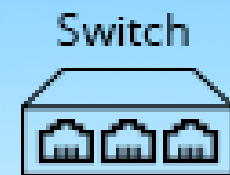
Storage/Data



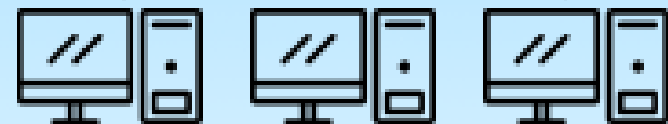
Router



END
USER



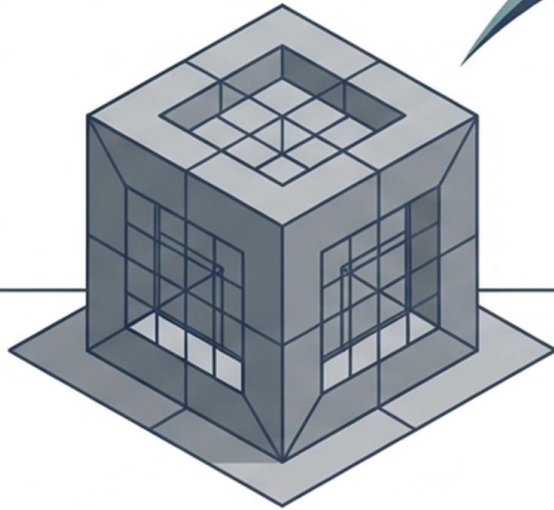
Switch



IT: FROM FIXED ASSETS TO ON-DEMAND SERVICES

On-Premise Model

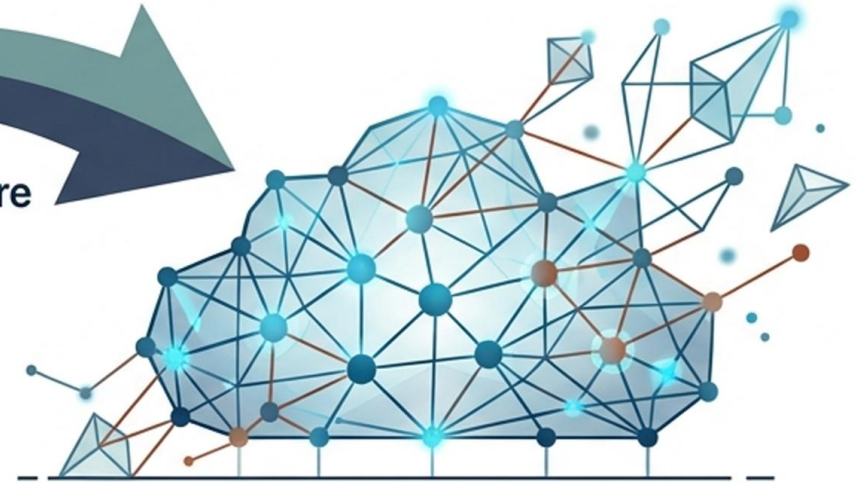
- Capital Expenditure (CapEx)
- Firm owns/manages hardware
- Fixed capacity (Hard to scale)
- High barrier to entry



A company keeps all of IT environment (locally) **on-site** managed either by itself or by a third-party.

The Infrastructure Shift

Cloud Model (SaaS/IaaS)



- Operating Expenditure (OpEx)
- Pay-as-you-go pricing
- Infinite scalability
- Accessed via Internet (AWS, Azure, Google)

The delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—**over the Internet (“the cloud”)**

ON-PREMISES COMPUTING: IT AS FIXED ASSETS

THE "MADE IN-HOUSE" MODEL: CONTROL VS. RIGIDITY

THE FIRM PURCHASES, INSTALLS, AND MANAGES ITS OWN PHYSICAL HARDWARE AND SOFTWARE WITHIN ITS OWN DATA CENTERS.

Benefits (The Power of Control)

Full Internal Control: Total authority over systems, configurations, and data.

Security Ownership: Direct management of data sovereignty and strict regulatory compliance.

Customization: Ability to build highly specific environments tailored to unique needs.

Drawbacks (The Burden of Ownership)

High Upfront Investment (CapEx): Massive capital required before any value is realized.

Fixed Capacity: Rigid infrastructure. Leads to wasted resources (if over) or lost revenue (if under).

Operational Burden: Heavy internal IT load for constant maintenance, patching, and hardware refreshes.

CLOUD COMPUTING: IT AS ON-DEMAND SERVICES

THE “ON-DEMAND SERVICE” MODEL: FLEXIBILITY & SCALABILITY

ACCESSING COMPUTING RESOURCES (SERVERS, STORAGE, SOFTWARE) OVER THE INTERNET ON A PAY-AS-YOU-GO BASIS, TREATING IT AS A UTILITY RATHER THAN A FIXED ASSET.

Benefits (The Power of Elasticity)

Scalability (Elasticity): Rapidly expand or shrink resources to match demand; converts CapEx to OpEx.

Speed and Agility: New systems can be deployed in minutes, enabling rapid experimentation and innovation.

Global Reach & Resilience: Deploy applications worldwide easily; "Built-in Backups" to keep services running even if parts fail.

Drawbacks (The Burden of Reliance)

Loss of Control: Trading off direct management and visibility for speed and flexibility.

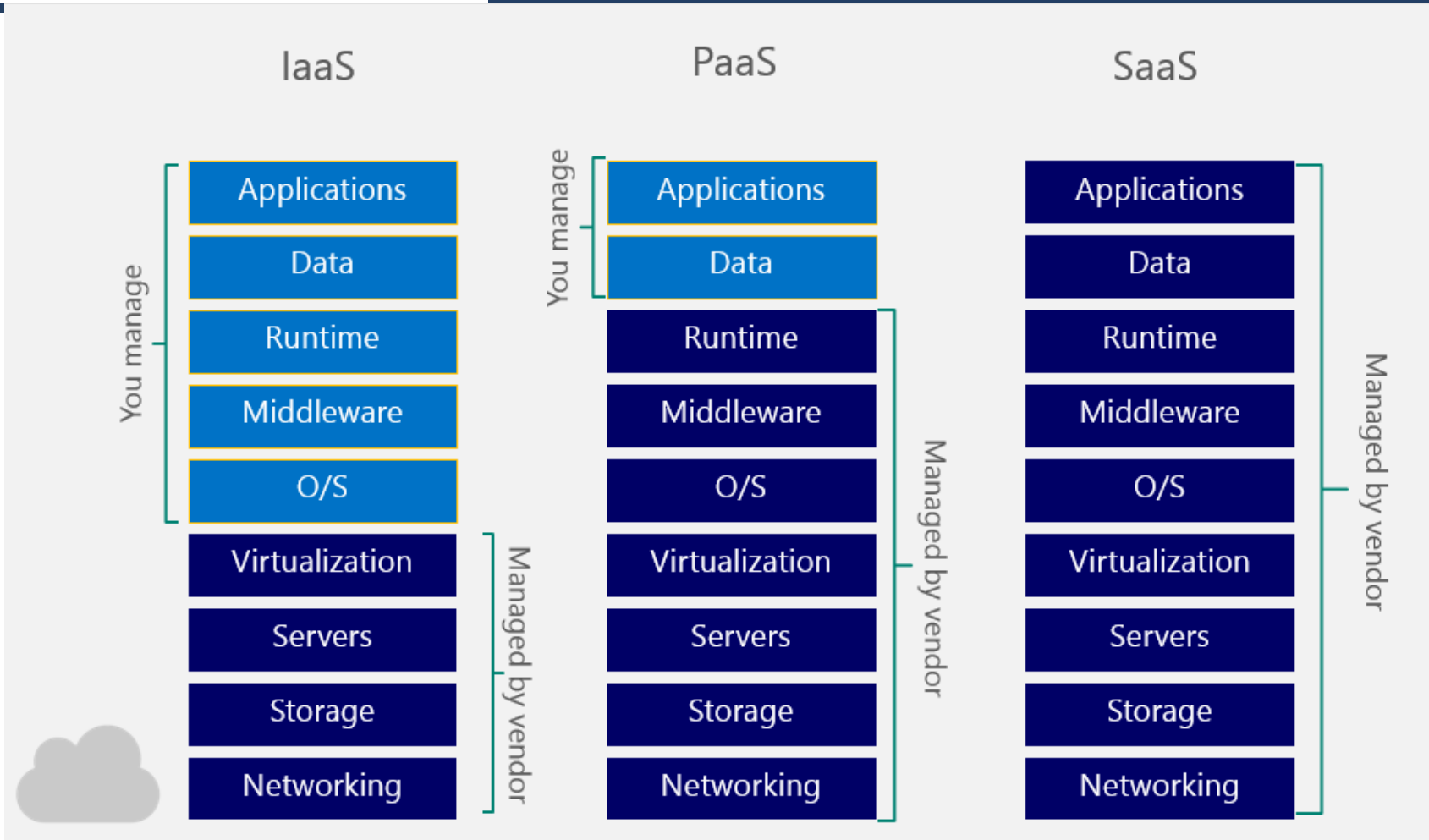
Vendor Lock-in: High switching costs and technical dependence on major providers (AWS, Azure, Google).

Security & Compliance: Data privacy concerns and complex regulations when hosting with third parties.

Infrastructure as a service

Platform as a service

Software as a service

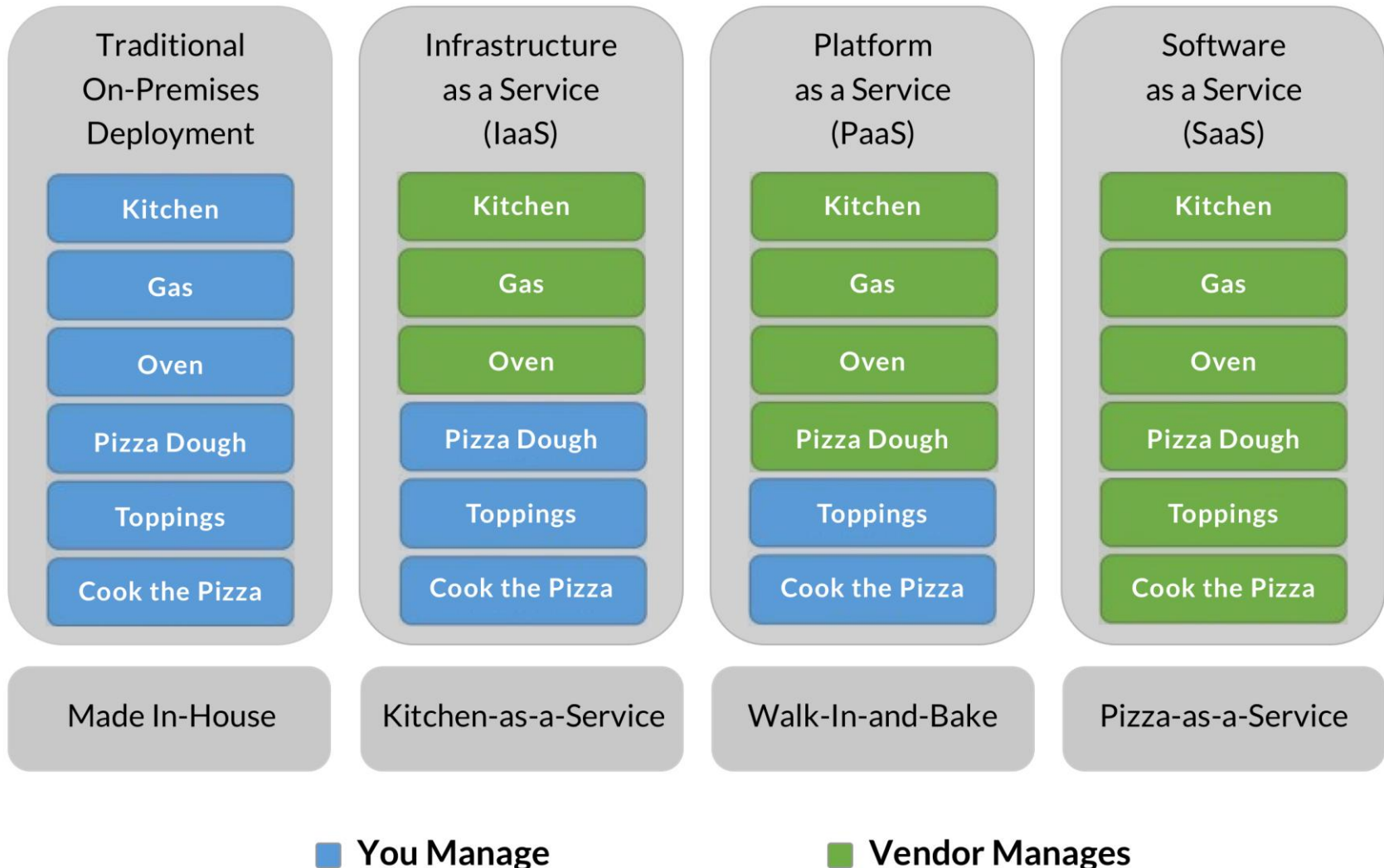


- Pre-configured HW
- OS needed
- Apps needed

- Operating environment included
- E.g, Windows / .NET, Linux

- Operating environment largely irrelevant
- Fully functional apps provided (e.g., CRM, ERP, email)

New Pizza as a Service



CLOUD COMPUTING MODEL COMPARISON

Feature	IaaS (Infrastructure)	PaaS (Platform)	SaaS (Software)
Technical Focus	Fundamental Resources: Virtual servers, storage, & networking.	Development Platform: Managing apps without infra complexity.	Finished Software: Fully owned and managed by a provider.
Concept Analogy	"Kitchen-as-a-Service"	"Walk-In-and-Bake"	"Pizza-as-a-Service"
Provider Manages	Kitchen, Gas, and Oven (Hardware & Virtualization).	Above + Dough & Kneading (OS, Middleware, & Runtime).	Above + Toppings & Baking (Data, Apps, and UI).
Your Responsibility	Making the Dough & Kneading, Toppings, and Baking.	Adding Toppings and Baking the Pizza.	None. You just use the service.
Key Phrase	"I have the kitchen; I need to knead the dough and build the rest."	"The dough is already kneaded; I'll just focus on my toppings."	"I don't cook; I just consume."
IT Layer Insight	You manage the OS and Runtime environment yourself.	The Runtime is pre-configured; you only manage Code/Data.	Everything is managed; you only manage your User Account.
Real-World Examples	Amazon EC2, Azure VM, Google Compute Engine (GCE).	Heroku, AWS Elastic Beanstalk, Google App Engine.	Salesforce, Slack, Gmail, Zoom, Microsoft 365.

IAAS VS PAAS VS SAAS: WHICH CLOUD SERVICE IS SUITABLE FOR YOU?

- If your business needs out-of-the-box software (CRM, email, collaboration tools, etc.), choose Software as a Service (SaaS).
- If your company requires a platform for building software products, pick Platform as a Service (PaaS).
- If your business needs a virtual machine (that provides basic computing infrastructure such as servers, storage, and networking resources), opt for Infrastructure as a Service (IaaS).

THE EVOLUTION TO AGENTIC SAAS

From Software as a Tool to Software as a Worker

- **Traditional SaaS:** A platform that requires human intervention to input data, navigate menus, and execute workflows.
- **Agentic SaaS:** A system where AI agents autonomously perform tasks and make decisions to achieve goals set by the user.

Feature	Traditional SaaS	Agentic SaaS
User Interaction	Manual navigation through GUIs (clicking, typing).	Goal-setting via Natural Language or APIs.
Primary Value	Providing tools for human efficiency.	Providing autonomous results and speed.
Pricing Model	Per-user "Seat" licenses (Scale by headcount).	Per-outcome or usage-based (Scale by value).
Responsibility	Human manages the process & data entry.	AI manages the workflow & execution.